SYSDESIGN REFERENCE

# Database Sharding

Core concepts, reference diagram, tradeoffs, and interview prompts.

Partition data so storage and throughput scale beyond one database boundary.

VERSION 1.0 REVIEW
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STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / DATABASE-SHARDING

SYSDESIGN

# Database Sharding

**CORE IDEA**

**Partition data so storage and throughput scale beyond one database boundary.**

## Reference architecture decision chain

| 01 | 02 | 03 | 04 |
|---|---|---|---|
| Shard Key | Routing | Rebalancing | Cross-Shard Queries |

### Decisions to defend

+ Shard key
+ routing
+ rebalancing
+ cross-shard queries
+ and tenant isolation.

### Failure modes to test

+ Hot shards
+ skew
+ resharding
+ and distributed transactions.

## Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

# Database Sharding

| Probe | What a strong answer makes explicit |
| --- | --- |
| How does the design behave during a regional dependency failure? | Requirement, decision, cost, failure behavior, and observable metric. |
| Which metric reveals the first bottleneck? | Requirement, decision, cost, failure behavior, and observable metric. |
| What consistency does the user actually observe? | Requirement, decision, cost, failure behavior, and observable metric. |
| How would the design change at ten times the traffic? | Requirement, decision, cost, failure behavior, and observable metric. |

## Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

## Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

**DESIGN DISCIPLINE**

**For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?**

PUBLISHING STATUS

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